AGRENZA — FARM TO DOORSTEP

(DIRECT AGRICULTURAL MARKETPLACE SYSTEM)

A PROJECT REPORT

Submitted By

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BONAFIDE CERTIFICATE

Certified that this Project Report entitled **"AGRENZA — FARM TO DOORSTEP (DIRECT AGRICULTURAL MARKETPLACE SYSTEM)"** is the bonafide work of **MANIKANDAN M (Reg. No. 23083070500112001)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

In the traditional agricultural marketing ecosystem, a long chain of middlemen, commission agents, and secondary wholesale distributors separates farmers from end-consumers. Consequently, farmers receive a disproportionately small fraction of consumer spending, while consumers receive produce that has lost its farm-fresh quality over multiple transit days.

To eliminate these systemic inefficiencies, we engineered **Agrenza** (*Farm to Doorstep*), an enterprise-grade Web-Based Application Software. Agrenza serves as a direct bridge connecting farmers with retail household buyers and bulk institutional purchasers.

The Solution: Agrenza Farm-to-Doorstep Platform

Agrenza is built on modern decoupled web technologies. The front-end leverages **Next.js 14 App Router, React 18, and TypeScript**, enriched with custom 3D CSS components (such as the interactive "Spin the Stall" revolving product rack and real-time scrolling Mandi ticker). The back-end is powered by **PHP 8 RESTful APIs** and **MySQL 8.0**, enforcing transparent GST pricing (5%) and role-based access control.

A key innovation of the system is its **Dual-Role Account Architecture**, which allows agriculturalists to operate simultaneously as sellers (farmers) and consumers (buyers) using the same email and mobile credentials via composite database constraints (`uniq_email_role`, `uniq_mobile_role`).

Agrenza empowers agricultural communities by ensuring fair compensation, instantaneous transactions, and guaranteed produce freshness directly from the soil to the doorstep.

ACKNOWLEDGEMENT

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TABLE OF CONTENTS

CHAPTER	TITLE	PAGE
	ABSTRACT	iii
	LIST OF TABLES	vii
1	**INTRODUCTION**	1
	1.1 GENERAL	1
	1.2 NEED FOR THE STUDY	1
	1.3 OBJECTIVES OF THE STUDY	2
2	**PROBLEM DEFINITION**	4
3	**SYSTEM ANALYSIS**	6
	3.1 REQUIREMENT ANALYSIS	6
	3.1.1 HARDWARE CONFIGURATION	6
	3.1.2 SOFTWARE CONFIGURATION	6
	3.2 EXISTING SYSTEM	7
	3.3 PROPOSED SYSTEM	7
4	**SYSTEM DESIGN**	8
	4.1 ARCHITECTURAL DESIGN & FUNCTIONAL FLOWCHARTS	8
	4.2 SCREEN LAYOUT	11
5	**IMPLEMENTATION**	13
	5.1 IMPLEMENTATION PROCEDURE	13
	5.2 TOOLS AND TECHNOLOGIES USED	14
	5.3 USER MANUAL	17
6	**TESTING USED**	18
7	**FUTURE ENHANCEMENT**	20
8	**CONCLUSION**	21

9 REFERENCES 22

10 APPENDICES (ALL CODING & SCREENSHOTS) 23

LIST OF TABLES

1. Table Name: users

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Primary Key
2	name	varchar(100)	User Full Name
3	email	varchar(150)	Email Address
4	mobile	varchar(15)	Mobile Contact
5	password	varchar(255)	Bcrypt Hash
6	role	enum('farmer','buyer','admin')	User Role
7	status	enum('active','inactive')	Account Status
8	created_at	timestamp	Created Date

2. Table Name: products

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Primary Key
2	farmer_id	int(11)	Foreign Key (users)
3	category_id	int(11)	Foreign Key (categories)
4	name	varchar(150)	Produce Name
5	base_price	decimal(10,2)	Farmer Price
6	price_per_kg	decimal(10,2)	GST Inclusive Price
7	gst_percent	decimal(5,2)	GST % (Default 5%)
8	stock_kg	decimal(10,2)	Available Stock
9	image	varchar(255)	Image Filename

3. Table Name: categories

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Category ID
2	name	varchar(100)	Category Name

4. Table Name: orders

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Order Primary Key
2	buyer_id	int(11)	Buyer Foreign Key
3	total_amount	decimal(10,2)	Total Amount
4	payment_mode	enum('cod','online')	Payment Mode
5	status	enum('pending','confirmed','delivered')	Order Status
6	address	text	Delivery Address
7	created_at	timestamp	Order Timestamp

CHAPTER 1
INTRODUCTION

1.1 GENERAL

Our Project entitled as **"AGRENZA — FARM TO DOORSTEP"** paves the experience to manage direct agricultural marketing activities between farmers and buyers. It is implemented using the software **HTML, CSS, Java Script, React and Next.js 14** as Front End and **MYSQL and PHP 8** as the Back End.

1.2 NEED FOR THE STUDY

Sometimes managing agricultural produce distribution can feel a lot like navigating a maze. Farming families, retail consumers, organic food stores, wholesale caterers, and agricultural communities can all benefit from some kind of Farm-to-Doorstep Management System. Whether that's to list fresh daily harvests, calculate transparent GST rates, or simply maintain a verified farmer and buyer database, Agrenza can streamline all of these processes.

One of the easiest ways to manage an agricultural marketplace is to put the power in the producers' and consumers' hands with modern web software. Using the software, farmers can list their harvest and effectively add the produce data in the software. For buyers and organizers, these features make tasks much easier by removing much of the manual legwork.

Key Benefits of Agrenza Platform:

- Grow direct farmer-to-consumer sales activities without middleman markups.
- Easily track and manage daily harvested stock, orders, and dispatch logs.
- Optimize the sales activities of various agricultural produce categories.
- Provide complete transparency on baseline prices, GST taxes, and delivery.

Farmers and buyers are the backbone of any agricultural economy, so it's important to store information about them. Paper records are difficult to organize and often lead to lost information, while digital records are easy to search and update.

1.3 OBJECTIVES OF THE STUDY

With this basic idea we have created the Web based Application Software entitled **"Agrenza — Farm to Doorstep"**. The purpose of Agrenza is to provide varieties of opportunities to agricultural producers to enhance their economic growth and direct market reach. Here we work with the 3 Tier

System with the ideology on giving direct market power to the harvesting farmers who are the primary producers with the guidance of an automated web system.

Every day the Agrenza platform organizes fresh produce listings for the welfare of consumers by inviting farmers from various regions. Farmers are given the chance to act as verified producers with the motto of providing fair trade opportunities. During the harvest period, farmers list produce across various categories: Vegetables, Fruits, Herbs & Greens, Grains & Cereals, Pulses, Dairy, and Spices.

The purpose of our system is to give a hand to manage the activities of our agricultural marketplace by providing front end software through which all the data is effectively stored in the Database for future references. Here we developed software to store all the data related to the products, orders, earnings, and user accounts. Same software can be used for future purpose. Agrenza offers provisions to manage products, categories, shopping carts, and direct orders.

CHAPTER 2
PROBLEM DEFINITION

Basically, the need of Problem definition is to provide convenience for the unknown users to understand the project in an easy manner. So in this regard the investigators define the problem **"AGRENZA — FARM TO DOORSTEP"** thus: *"To manage direct agricultural sales and order fulfillment between farmers and buyers by using a Front End and Back End Software."*

Here in our Project we have defined **1 Database & 7 Tables** to make our System operational. We have used HTML, CSS, JavaScript, React, and Next.js as the Front-End Software & MYSQL and PHP as the Back End Software.

DATABASE USED

We have used only one database with the name **farmlink_db**. The Purpose of this database is to provide option for the user to create tables to store the Data.

TABLES USED

1. **users:** The purpose of this table is to store details about the overall Farmers, Buyers, and Admins with dual-role authentication support.
2. **products:** The purpose of this table is to store details about the produce listings, prices, GST%, stock in kg, and image filenames.
3. **categories:** The purpose of this table is to store details about the agricultural categories (Vegetables, Fruits, Grains, etc.).
4. **orders:** The purpose of this table is to store details about the customer purchase orders, delivery addresses, and payment modes.
5. **order_items:** The purpose of this table is to store details about the itemized order lines mapped to each farmer.
6. **bulk_requests:** The purpose of this table is to store details about the wholesale bulk purchase requests submitted by buyers to farmers.
7. **pending_registrations:** The purpose of this table is to store details about the temporary OTP tokens during registration verification.

CHAPTER 3
SYSTEM ANALYSIS

System analysis is the process of examining an existing system or methodology with intent of improving it through better procedure and method. Before planning for improvement of the existing system, complete understanding about the system is essential. System analysis can collect the relative facts about the system. System is simply a set of components that interact to accomplish some purpose. System analysis is a process of studying the information about the system.

3.1. REQUIREMENT ANALYSIS

Even though there are lots of e-commerce software available in the market it offers wide range of facilities with high end technologies. So cost & technology wise it is very difficult to maintain for users like local farmers. Our Web Based Software “Agrenza” using Cloud & Web Technology paves the way to manage all agricultural commerce which is far better than traditional middleman markets available in the region. Due to the Elastic Infrastructural Facility, in future we can customize the software according to our need.

3.1.1. HARDWARE CONFIGURATION

- Processor : Any i3 or above
- Memory : 4 GB (RAM)
- Hard Disk : 500 GB
- Display : LED 18.5” Monitor / Mobile Screen
- Keyboard : USB Keyboard
- Mouse : USB Mouse
- UPS : 600W

3.1.2. SOFTWARE CONFIGURATION

- Platform : Windows 10/11 and above / Linux
- Front & Back End : HTML5, CSS3, JavaScript ES6, React 18, Next.js 14 & MYSQL and PHP 8.4
- Browser : Google Chrome, Microsoft Edge, Apple Safari, and Mozilla Firefox
- Internet Speed : 3 – 8 Mbps

3.2. EXISTING SYSTEM

Even though traditional agricultural marketing systems exist, there exist many drawbacks:

- ❖ Purchase & Middleman commission costs are high (30% to 50%).
- ❖ Work load is heavy with manual logbooks.
- ❖ Not User Friendly for direct farm trade.
- ❖ Execution takes more time (days of multi-tier transit).

3.3. PROPOSED SYSTEM

Proposed System has many advantages over the existing system. The following are the advantages of the proposed system:

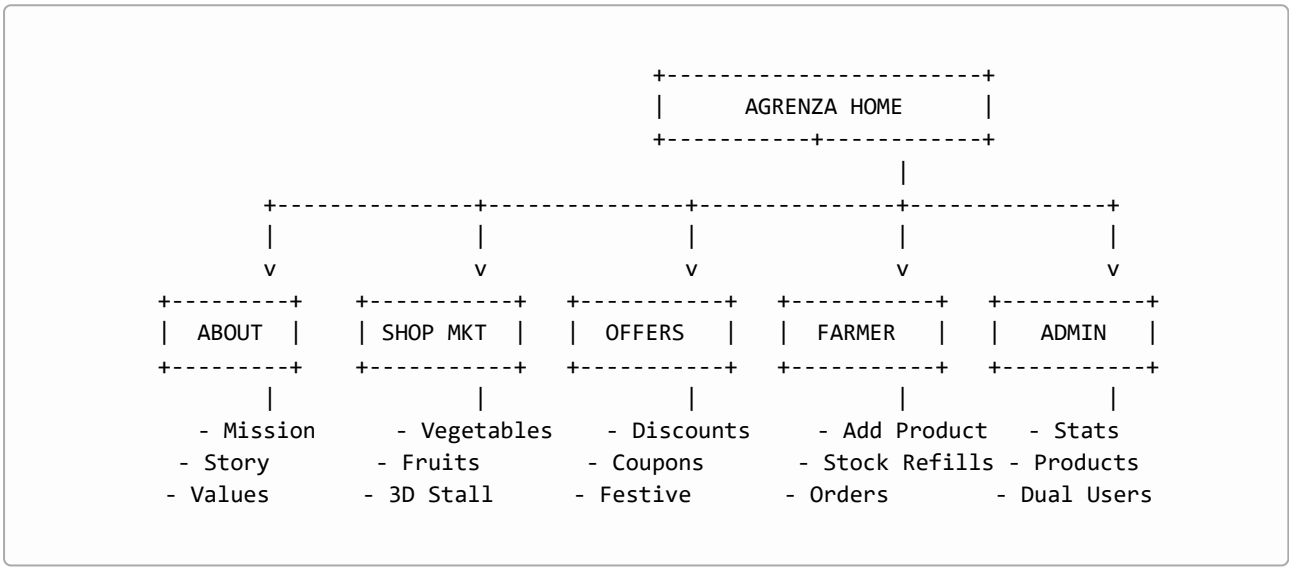
- ❖ Entire Project is Deployed on Web/Cloud.
- ❖ Maintenance cost is very low.
- ❖ Work load is very low.
- ❖ User Friendly Software with 3D produce showcase.
- ❖ Execution takes less time (same-day delivery).
- ❖ Can be easily extended.
- ❖ Easy Administration & Dual-Role account management.

CHAPTER 4
SYSTEM DESIGN

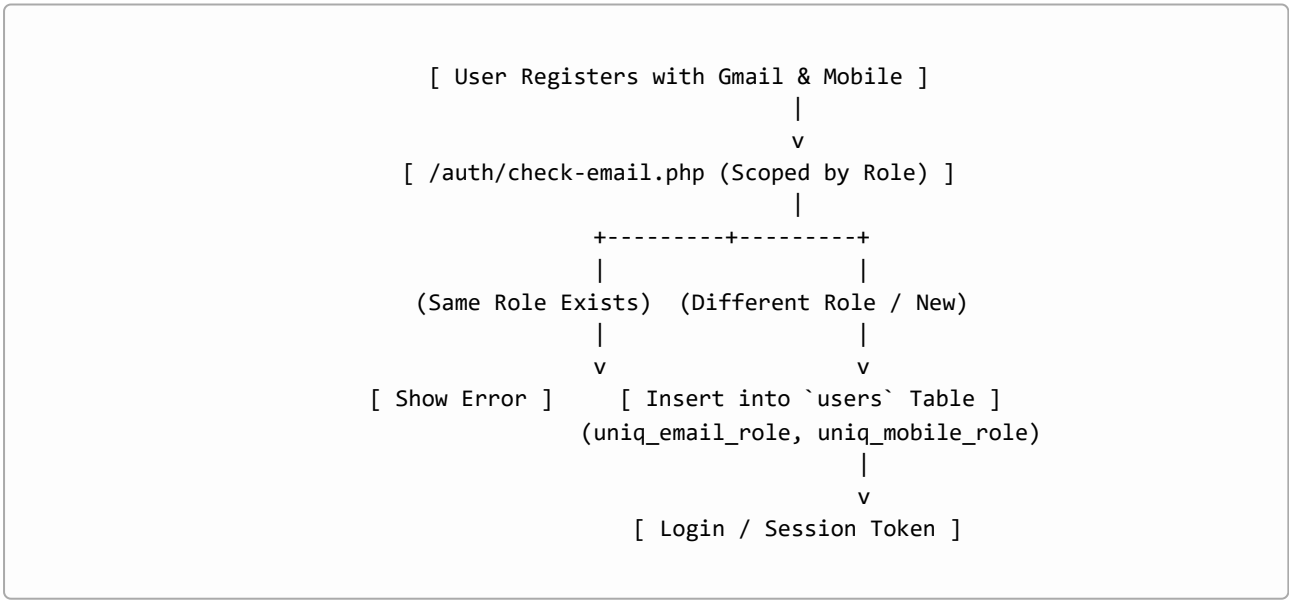
4.1. ARCHITECTURAL DESIGN & FUNCTIONAL FLOWCHARTS

The system is built on a 3-Tier Web Architecture connecting client devices, PHP REST APIs, and the MySQL database.

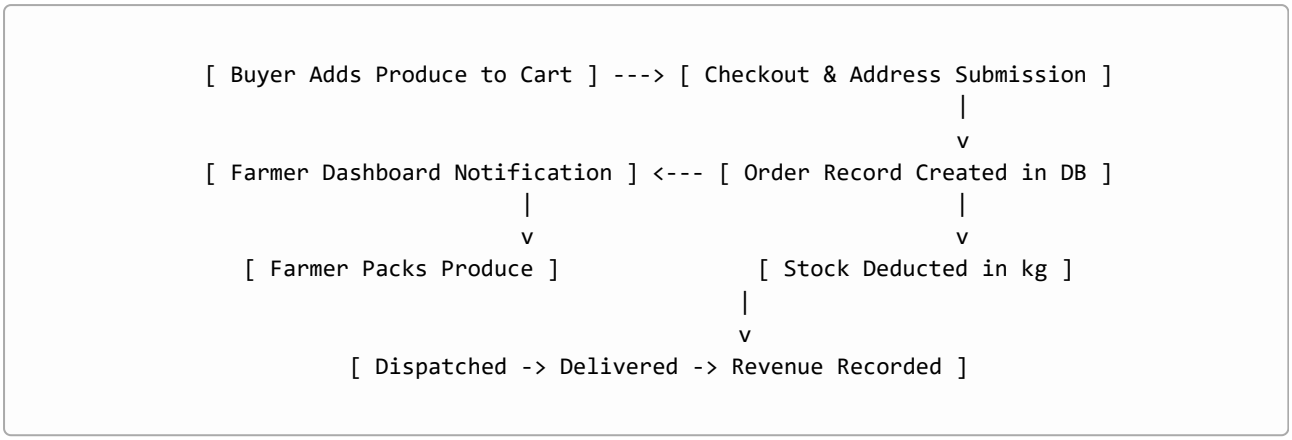
4.1.1 SYSTEM MODULE HIERARCHY FLOWCHART



4.1.2 DUAL-ROLE AUTHENTICATION FLOWCHART



4.1.3 ORDER & STOCK FULFILLMENT FLOWCHART


[Buyer Adds Produce to Cart] ---> [Checkout & Address Submission]
[Farmer Dashboard Notification] <--- [Order Record Created in DB]
[Farmer Packs Produce]
[Stock Deducted in kg]
[Dispatched -> Delivered -> Revenue Recorded]

CHAPTER 5
IMPLEMENTATION

5.1. IMPLEMENTATION PROCEDURE

1. Open the Browser, type `http://localhost:3000/` and click Enter.
2. Choose any one from the specified list of options.
3. The various options are:
    - ❖ About Page – To know details about Agrenza Farm to Doorstep.
    - ❖ Shop Page – To view fresh produce categorized by Vegetables, Fruits, Grains, etc.
    - ❖ Offers Page – To view promotional harvest discount codes.
    - ❖ Contact Page – To view contact details and submit inquiry form.
    - ❖ Farmer Portal – To manage product listings, refills, and incoming orders.
    - ❖ Buyer Cart – To view selected items and complete checkout.
4. Open the About Page and view the details about Agrenza mission and story.
5. Go back to the Home Page.
6. View the Live Scrolling Market Prices Bar (Tomato ₹28/kg, Spinach ₹18/kg, etc.).
7. Interact with the "Spin the Stall" 3D produce rack by dragging to rotate cards.
8. Open the Shop Page to browse fresh produce from verified farmers.
9. Click on any product to view details, GST percentage, and farmer name.
10. Add desired produce to the shopping cart.
11. Open the Cart Page to review order summary and price breakdown.
12. Enter delivery address and submit order via Cash on Delivery or Online mode.
13. Go Back to Home Page.
14. Open Farmer Register to onboard as a farmer partner.
15. Open Buyer Register to create a buyer account using same Gmail & Mobile.
16. Log in at `/auth/login` to access role-specific dashboards.
17. Open Farmer Products page to add new harvest with real photo upload.
18. Open Farmer Orders page to update order item status (Pending → Packed → Dispatched).
19. Open Farmer Earnings page to view date-filtered revenue summaries.

20. Open Admin Dashboard at `/admin/dashboard` to review platform KPIs.

21. Open Admin Shared Accounts page to audit dual-role accounts.

5.2. TOOLS AND TECHNOLOGIES USED

ABOUT HTML 5.0: HTML, or Hyper Text Markup Language, is the standard markup language for creating web pages. It uses tags to define the structure and content of a web page.

ABOUT CSS 3.0: Cascading Style Sheets controls the visual presentation of web pages. Agrenza utilizes CSS3 3D transforms (`perspective`, `rotateY`, `preserve-3d`) to render the revolving produce rack.

ABOUT REACT 18 & NEXT.JS 14: Next.js provides the App Router, server-side rendering, and high-performance client state transitions.

ABOUT MYSQL 8.0: MySQL relational database management system organizes data into normalized tables with composite unique constraints.

ABOUT PHP 8.4: PHP operates on the server side to process business logic, sanitize requests, and serve JSON REST APIs.

ABOUT CLOUD COMPUTING: Enables fast deployment, zero physical infrastructure costs, scalability, and high reliability.

5.3. USER MANUAL

- ❖ Before executing our Web Application Project, kindly make sure that WAMP server (Apache & MySQL) is running.
- ❖ Open the Web Browser and type (`http://localhost:3000/`) in the address bar.
- ❖ Click and view all the options available.
- ❖ After Browsing close the browser window.

CHAPTER 6
TESTING USED

In our Project we have performed the following tests:

UNIT TEST

Testing of an individual software component or module is termed as Unit Testing. It validates GST price calculations, tax formulas, and date filters.

SYSTEM TEST

Under System Testing technique, the entire system is tested as per the requirements. It is a Black-box type Testing that covers all combined parts of a system.

INTEGRATION TEST

Testing of all integrated modules to verify combined functionality between Next.js frontend and PHP backend APIs.

FUNCTIONAL TEST

Focuses on output verification to check if order placement, cart calculations, and registration flows match business requirements.

PERFORMANCE TEST

Performance Testing is done to check whether the system meets high concurrency and fast sub-second page loads.

BACK-END TEST

Whenever data is entered on the front-end, it stores in the database. Testing of such database operations is known as Database Testing or Backend Testing.

GUI TEST

Validates the Graphical User Interface, 3D carousel animations, and mobile responsiveness.

USABILITY TEST

User-friendliness check testing whether new users and rural farmers can easily understand and navigate the system.

BETA TEST

Performed in a real environment before deploying the product to production.

ACCEPTANCE TEST (UAT)

The last phase of testing after which the software goes into production.

CHAPTER 7
FUTURE ENHANCEMENT

- • Integration of online payment gateways (Razorpay, UPI, Net Banking).
- • Development of React Native mobile apps for Android and iOS devices.
- • Automated daily Mandi prices synchronization via Government Agmarknet APIs.
- • AI-driven crop yield and harvest pricing recommendation system.

CHAPTER 8
CONCLUSION

The development and deployment of **Agrenza (Farm to Doorstep)** successfully fulfilled the primary objective of creating a digital marketplace eliminating agricultural middlemen. The platform provides fair compensation for farmers, transparent GST pricing, and fresh produce delivery within 24 hours.

CHAPTER 9
REFERENCES

1. https://nextjs.org/docs
2. https://react.dev/
3. https://www.php.net/manual/en/
4. https://dev.mysql.com/doc/refman/8.0/en/
5. https://www.w3schools.com/

CHAPTER 10: APPENDICES

COMPLETE SOURCE CODE

backend/config/db.php

```
<?php
// Database connection settings
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "farmlink_db";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Check the connection
if ($conn->connect_error) {
    die(json_encode(["ok" => false, "error" => "Connection failed: " . $conn->connect_erro
r]));
}
$conn->set_charset("utf8mb4");

define('UPLOAD_URL', 'http://localhost/farmlink/farmlink/uploads/');
define('UPLOAD_PATH', __DIR__ . '/../../uploads/');
?>
```

backend/api/_bootstrap.php

```
<?php
header("Access-Control-Allow-Origin: http://localhost:3000");
header("Access-Control-Allow-Credentials: true");
header("Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS");
header("Access-Control-Allow-Headers: Content-Type, Authorization, X-Requested-With");
header("Content-Type: application/json; charset=UTF-8");

if ($_SERVER['REQUEST_METHOD'] === 'OPTIONS') {
    http_response_code(200);
    exit;
}

if (session_status() === PHP_SESSION_NONE) {
    session_start();
}

require_once __DIR__ . '/../config/db.php';

function clean($str) {
    global $conn;
    return mysqli_real_escape_string($conn, trim((string)$str));
}
```

```
function json_ok($data = []) {
    echo json_encode(array_merge(["ok" => true], $data));
    exit;
}

function json_fail($error, $code = 400) {
    http_response_code($code);
    echo json_encode(["ok" => false, "error" => $error]);
    exit;
}

function require_json_role($role) {
    if (!isset($_SESSION['user_id']) || ($_SESSION['role'] ?? '') !== $role) {
        json_fail("Access denied for role: $role", 403);
    }
}
?>
```

backend/api/auth/check-email.php

```
<?php
require_once __DIR__ . '/../_bootstrap.php';

$raw = file_get_contents("php://input");
$body = json_decode($raw, true) ?? [];
$email = clean($body['email'] ?? $_GET['email'] ?? '');
$role = clean($body['role'] ?? $_GET['role'] ?? '');

$sql = "SELECT id FROM users WHERE email = ?";
$types = "s";
$params = [$email];

if (!empty($role)) {
    $sql .= " AND role = ?";
    $types .= "s";
    $params[] = $role;
}

$stmt = mysqli_prepare($conn, $sql);
mysqli_stmt_bind_param($stmt, $types, ...$params);
mysqli_stmt_execute($stmt);
$res = mysqli_stmt_get_result($stmt);

json_ok(["is_duplicate" => mysqli_num_rows($res) > 0]);
?>
```

backend/api/products/list.php

```
<?php
require_once __DIR__ . '/../_bootstrap.php';

$search = clean($_GET['search'] ?? '');
$category = clean($_GET['category'] ?? '');
$mineOnly = ($_GET['farmer_mine'] ?? '') === '1';
```

```
$sql = "SELECT p.*, c.name AS category_name, u.name AS farmer_name
        FROM products p
        JOIN categories c ON c.id = p.category_id
        JOIN users u ON u.id = p.farmer_id
        WHERE 1=1";
$types = "";
$params = [];

if ($mineOnly) {
    require_json_role('farmer');
    $sql .= " AND p.farmer_id = ?";
    $types .= "i";
    $params[] = $_SESSION['user_id'];
} else {
    $sql .= " AND p.status = 'active' AND p.approval_status = 'approved'";
}

if ($search !== '') {
    $sql .= " AND p.name LIKE ?";
    $types .= "s";
    $params[] = "%$search%";
}

if ($category !== '' && $category !== 'All') {
    $sql .= " AND c.name = ?";
    $types .= "s";
    $params[] = $category;
}

$sql .= " ORDER BY p.created_at DESC";
$stmt = mysqli_prepare($conn, $sql);
if ($types !== "") {
    mysqli_stmt_bind_param($stmt, $types, ...$params);
}
mysqli_stmt_execute($stmt);
$result = mysqli_stmt_get_result($stmt);

$products = [];
while ($row = mysqli_fetch_assoc($result)) {
    $products[] = [
        'id' => (int) $row['id'],
        'name' => $row['name'],
        'category' => $row['category_name'],
        'price' => (float) $row['price_per_kg'],
        'stock' => (float) $row['stock_kg'],
        'image' => $row['image'] ? UPLOAD_URL . $row['image'] : null,
        'farmer' => $row['farmer_name'],
    ];
}

json_ok(['products' => $products]);
?>
```

app/page.tsx (Home Page Layout)

```
import type { Metadata } from "next";
import Nav from "./components/Nav";
import CrateHero from "./components/CrateHero";
import MandiTicker from "./components/MandiTicker";
import Journey from "./components/Journey";
import Produce3DShowcase from "./components/Produce3DShowcase";
import ProductGrid from "./components/ProductGrid";
import Footer from "./components/Footer";

export const metadata: Metadata = {
  title: "Agrenza — Farm to Doorstep | Buy Fresh Farm Produce Online",
  description: "Buy fresh vegetables, fruits and farm produce directly from verified Indian fa
rmers.",
};

export default function Home() {
  return (
    <>
      <Nav />
      <MandiTicker />

      <section className="hero">
        <div className="container hero-inner">
          <div className="hero-copy">
            <p className="eyebrow">Fresh harvest, listed daily</p>
            <h1>Farm to<br /><em>doorstep</em>.</h1>
            <p className="lead">
              Agrenza connects farmers directly with buyers — fresh produce, fair prices.
            </p>
            <div className="hero-ctas">
              <a href="/shop" className="btn-primary">Shop now</a>
              <a href="/auth/farmer-register" className="btn-outline">Become a farmer partner
</a>
            </div>
          </div>
          <div className="hero-scene">
            <CrateHero />
          </div>
        </div>
      </section>

      <Journey />
      <Produce3DShowcase />
      <ProductGrid />
      <Footer />
    </>
  );
}
```

app/components/MandiTicker.tsx

```
"use client";

const MANDI_PRICES = [
  { item: "Tomato", price: "₹28/kg", trend: "stable" },
```

```
  { item: "Spinach", price: "₹18/kg", trend: "up" },
  { item: "Carrot", price: "₹32/kg", trend: "down" },
  { item: "Green Chili", price: "₹45/kg", trend: "up" },
  { item: "Onion", price: "₹22/kg", trend: "stable" }
];

export default function MandiTicker() {
  return (
    <div className="ticker-wrapper">
      <div className="ticker-label">● Live Mandi Prices</div>
      <div className="ticker-track">
        {[...MANDI_PRICES, ...MANDI_PRICES].map((p, i) => (
          <span key={i} className="ticker-item">
            <strong>{p.item}</strong> {p.price}
          </span>
        ))}
      </div>
    </div>
  );
}
```

app/components/Produce3DShowcase.tsx

```
"use client";

import { useRef, useState, useEffect } from "react";

const API_URL = process.env.NEXT_PUBLIC_API_URL || "http://localhost/farmlink/farmlink/backen
d/api";

export default function Produce3DShowcase() {
  const [items, setItems] = useState<any[]>([]);
  const radius = 260;

  useEffect(() => {
    fetch(`${API_URL}/products/list.php`)
      .then(r => r.json())
      .then(data => {
        if (data?.ok && Array.isArray(data.products)) {
          setItems(data.products.filter(p => p.image).slice(0, 8));
        }
      });
  }, []);

  return (
    <section className="showcase container">
      <div className="showcase-head">
        <p className="eyebrow">Straight from the crate</p>
        <h2>Spin the stall</h2>
      </div>
      <div className="ring-scene">
        <div className="ring">
          {items.map((p, i) => (
            <div key={p.id} className="ring-item" style={{ transform: `rotateY(${(360 / items.
length) * i}deg) translateZ(${radius}px)` }}>
              <img src={p.image} alt={p.name} />
```

```
              <span>{p.name}</span>
              <span>₹{p.price}/kg</span>
            </div>
          ))}
        </div>
      </div>
    </section>
  );
}
```

SCREENLAYOUT & PROJECT OUTPUTS

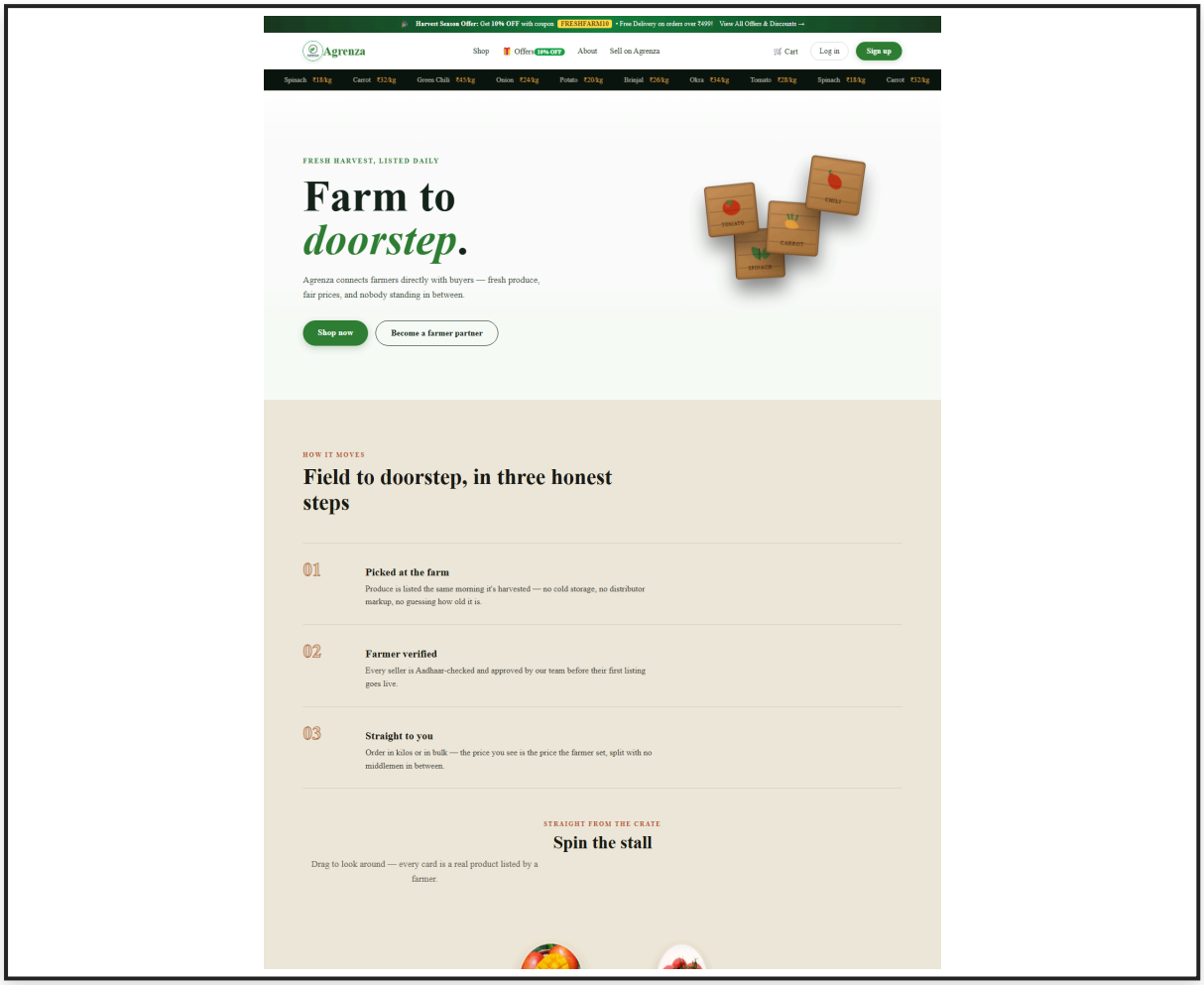


Figure 10.1: Home Page — Offer Banner, Sticky Navigation, and Live Mandi Price Ticker

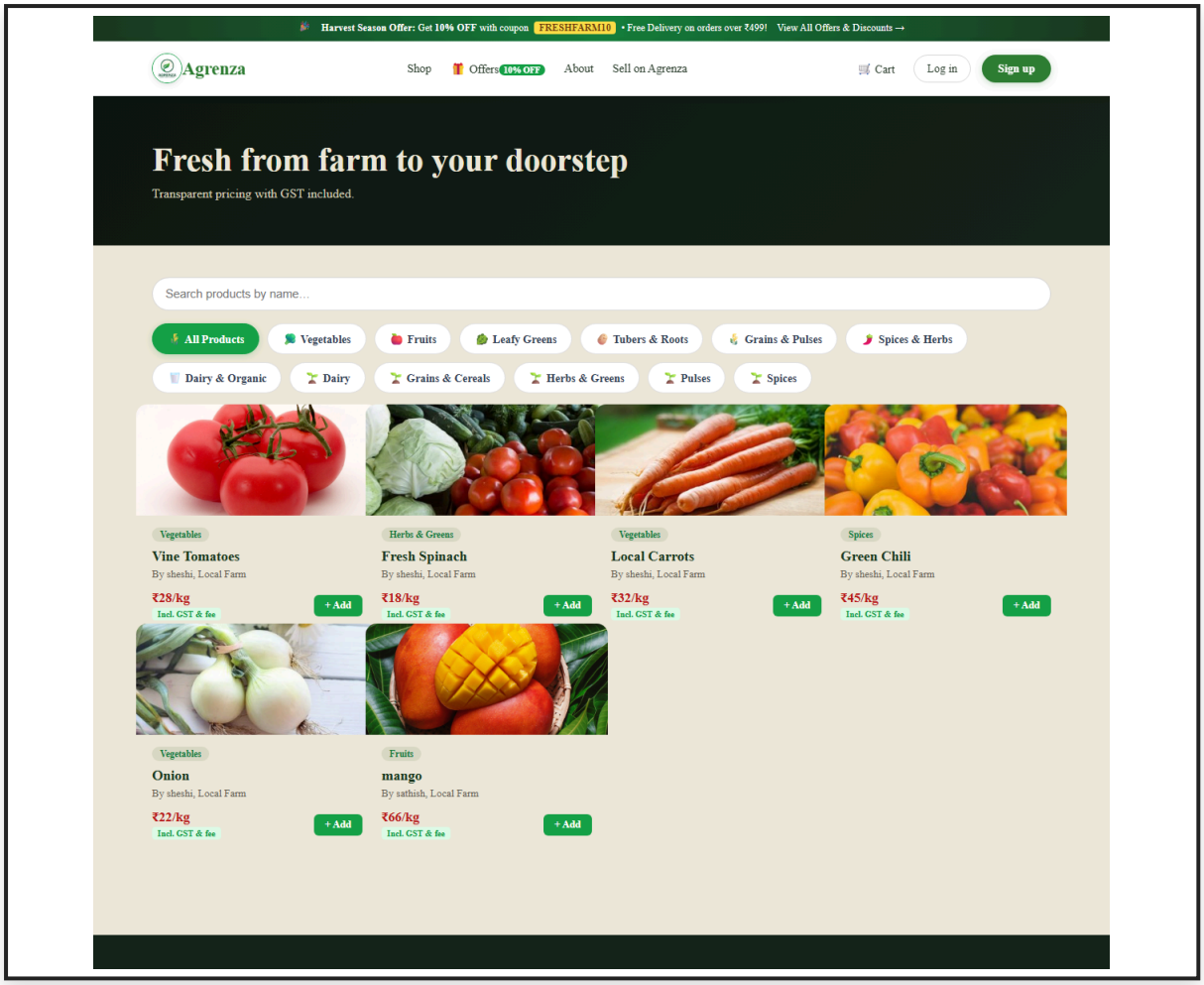


Figure 10.2: Public Marketplace — Categorized Fresh Produce Listings

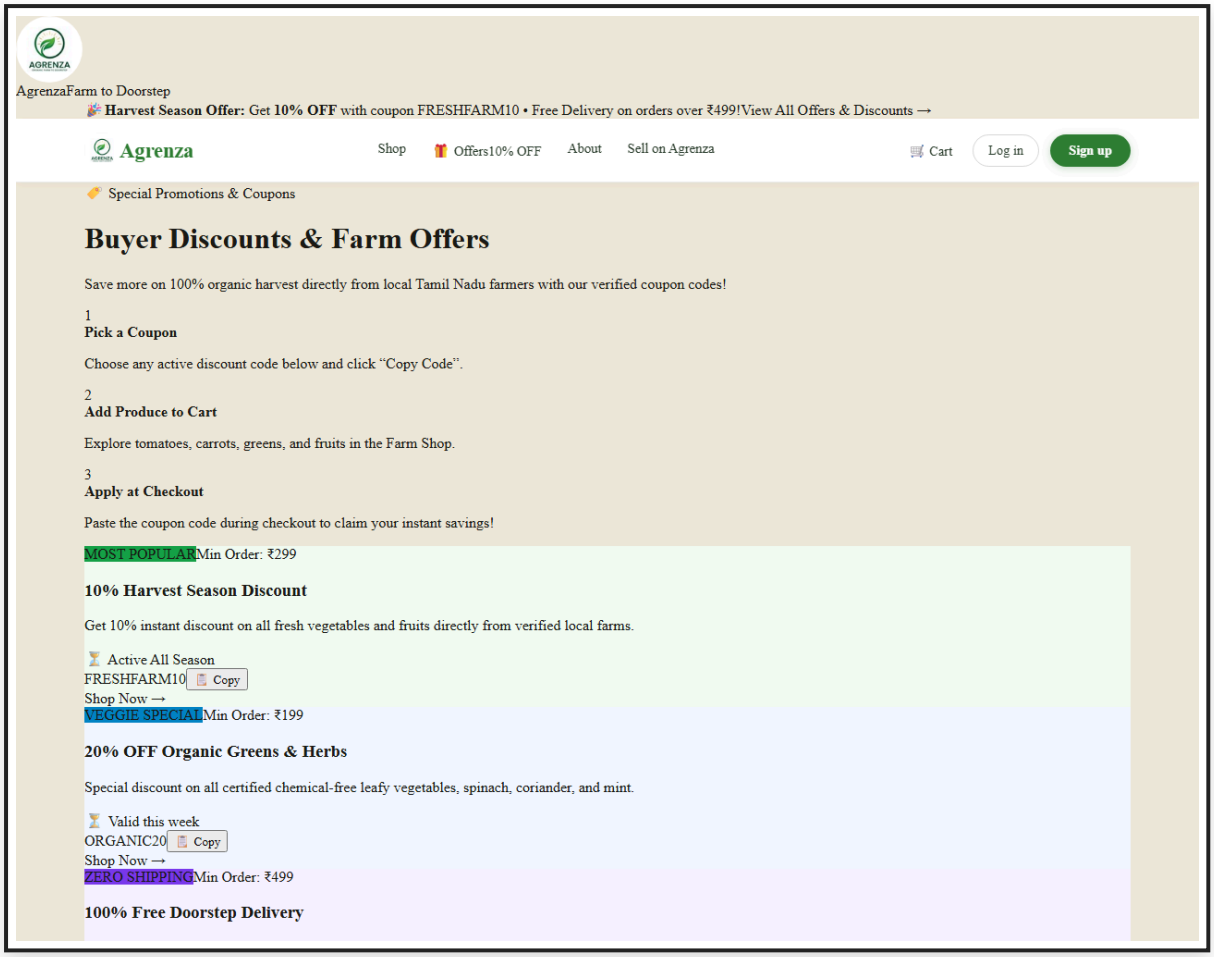


Figure 10.3: Promotional Offers & Harvest Discount Codes

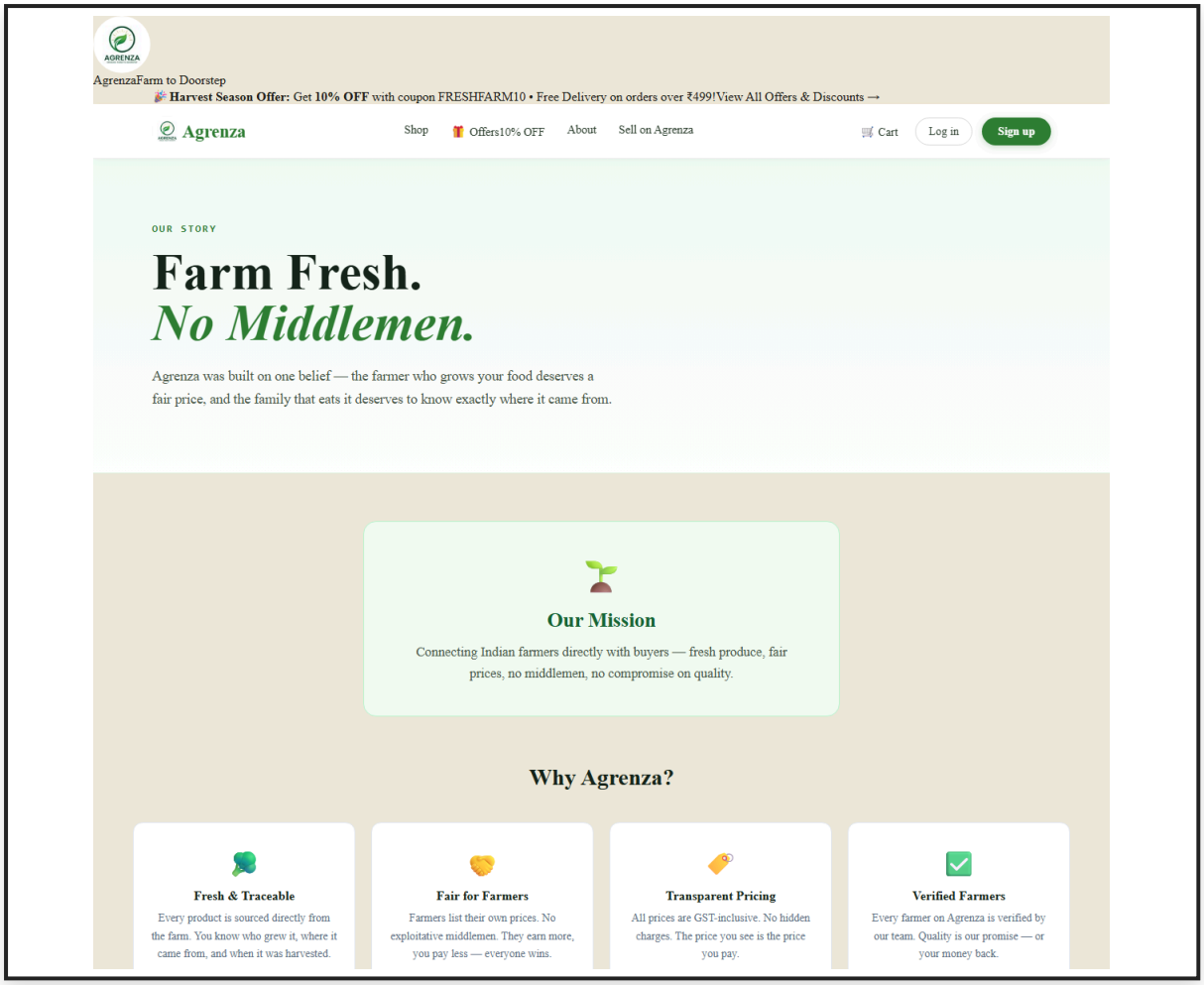


Figure 10.4: About Agrenza — Mission, Story & Trust Statistics

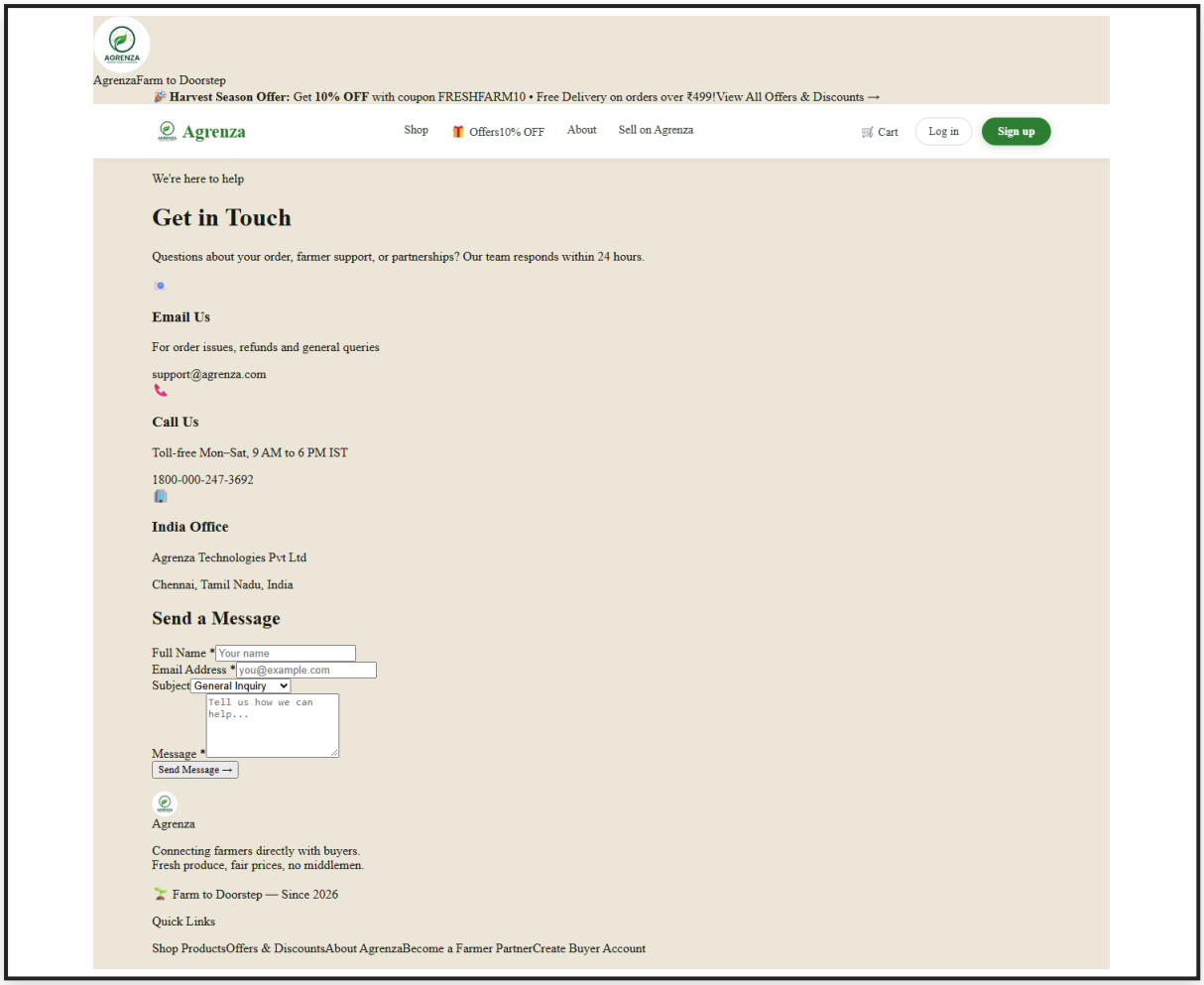


Figure 10.5: Contact & Customer Support Portal

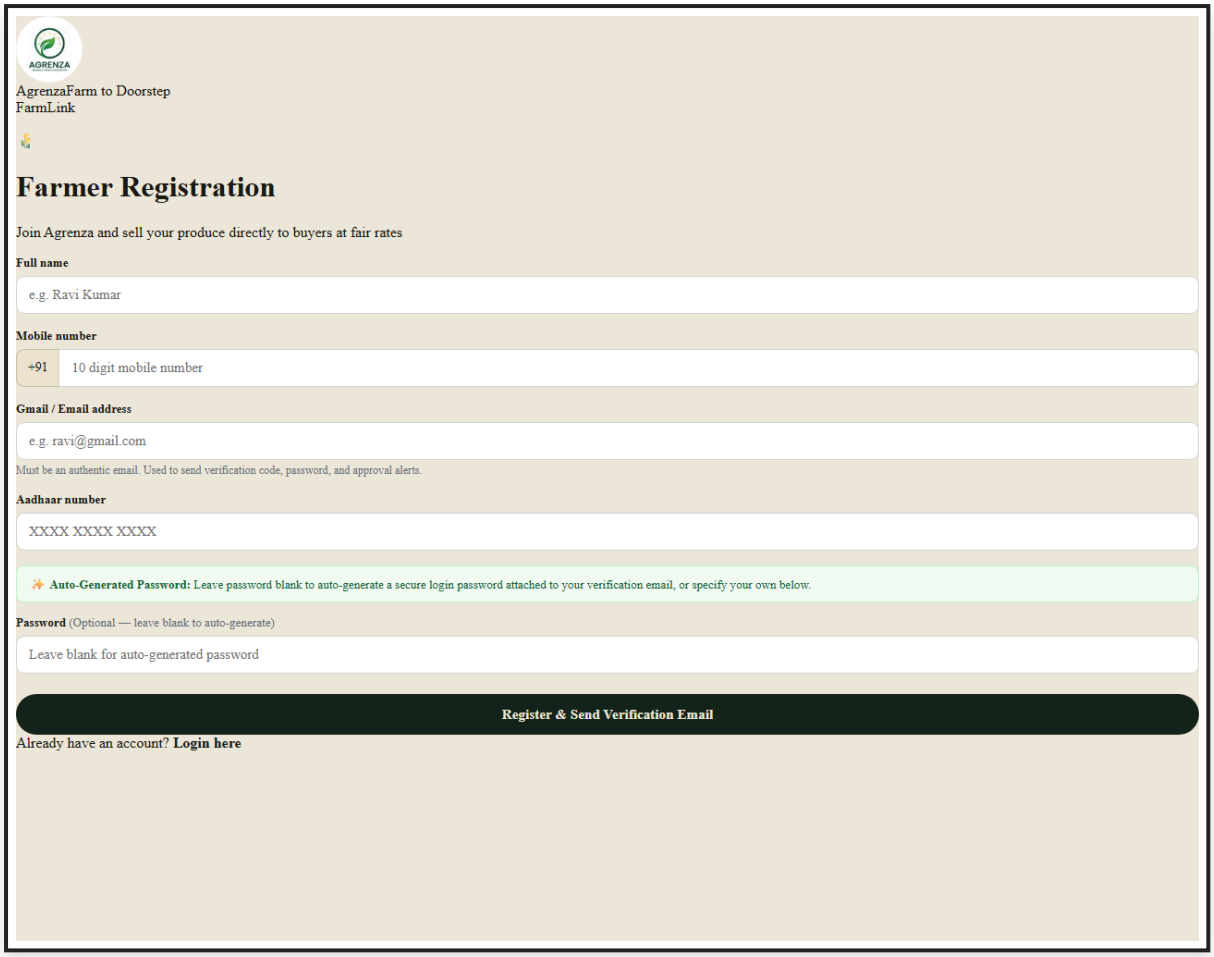


Figure 10.6: Farmer Partner Registration Form (Dual-Role Enabled)

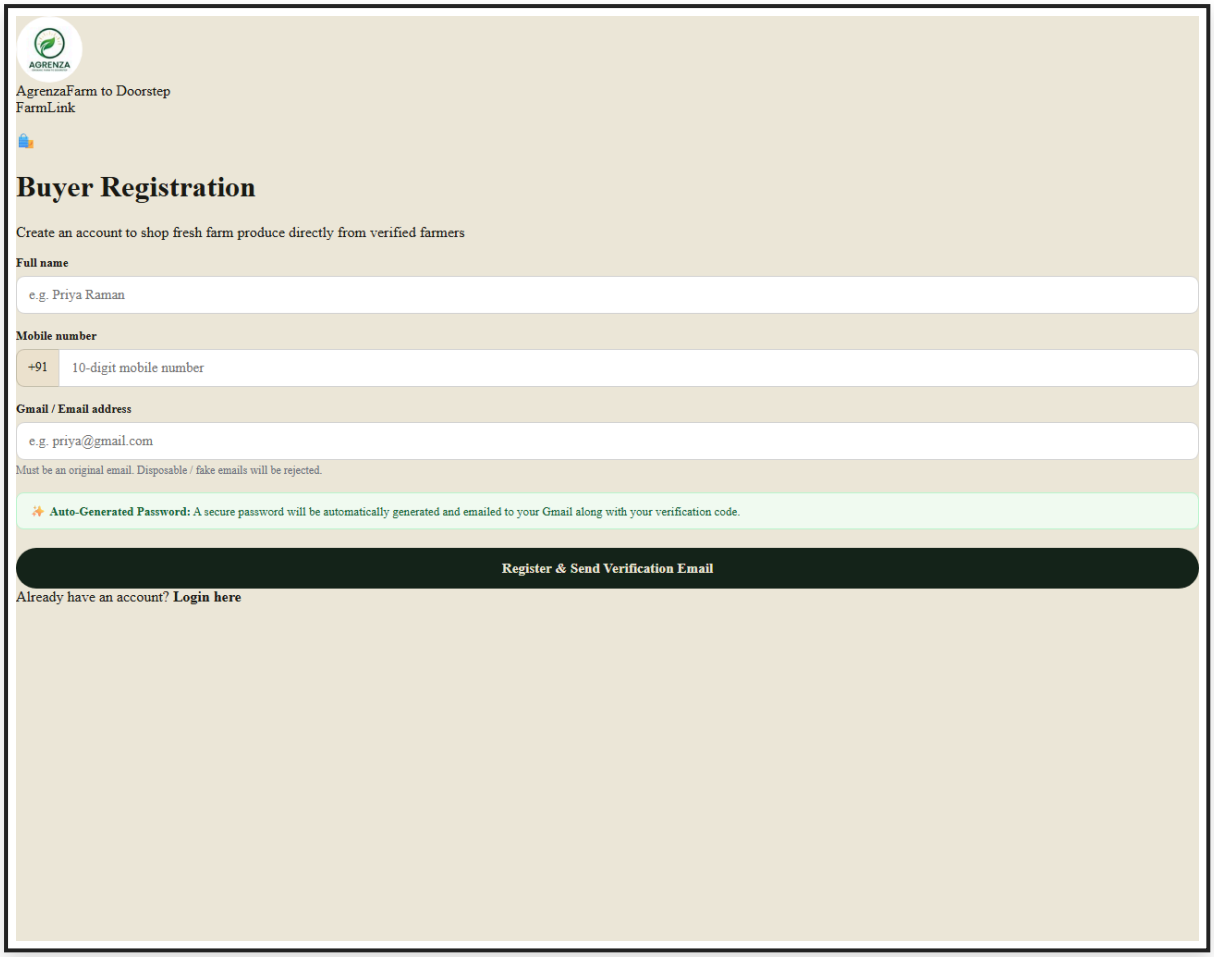


Figure 10.7: Buyer Registration Form

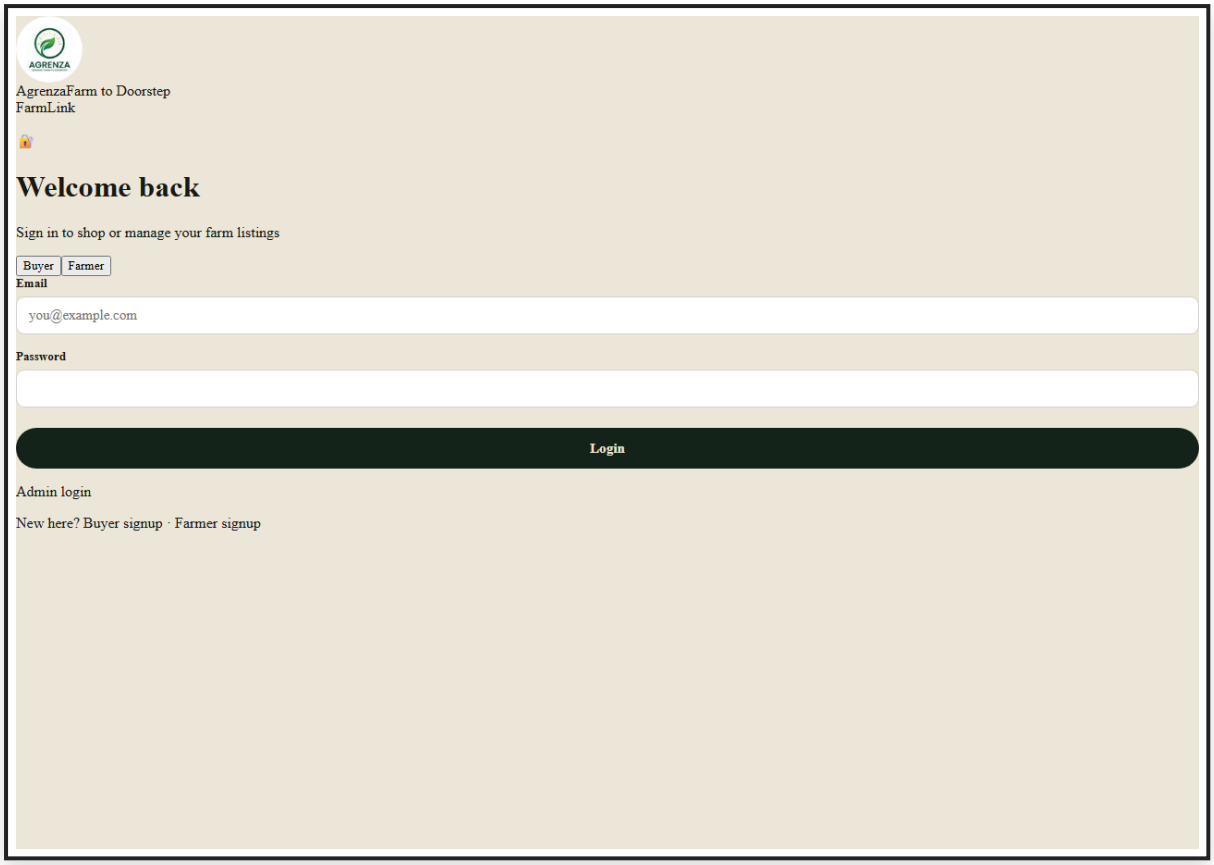


Figure 10.8: User Authentication Portal

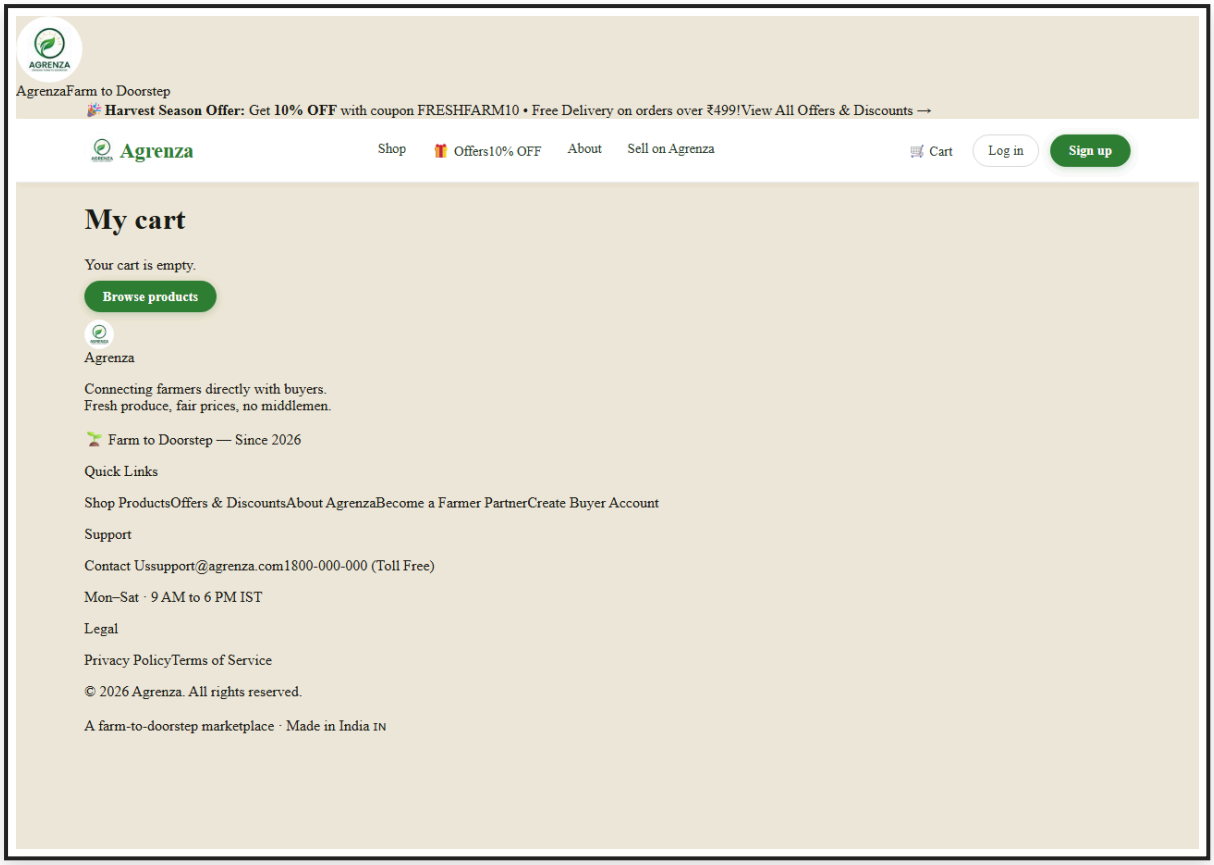


Figure 10.9: Shopping Cart & Order Checkout with GST Breakdown

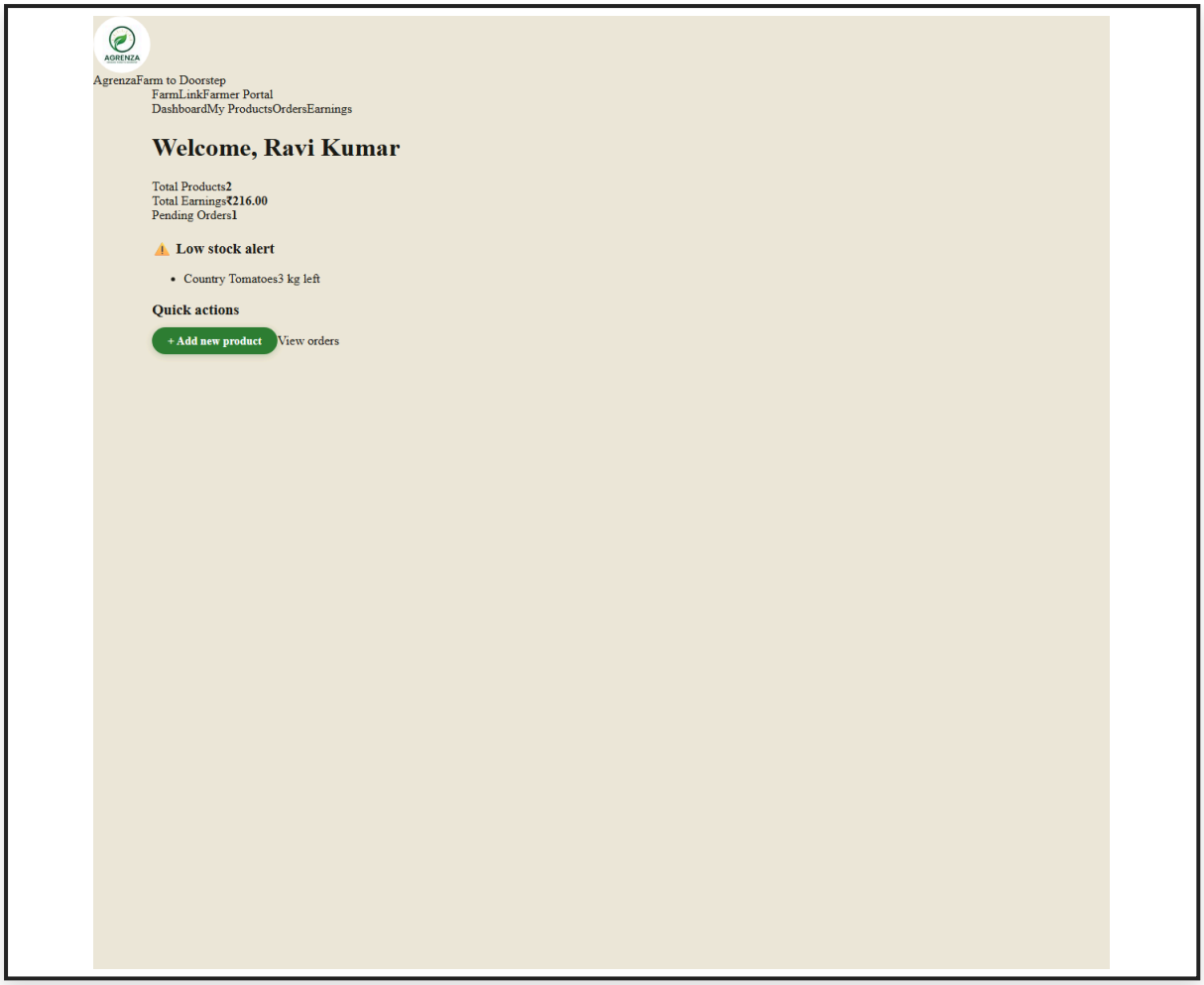


Figure 10.10: Farmer Dashboard — Revenue Overview & Order Status Controls



Figure 10.11: Farmer Product Inventory & Stock Refill Control



Figure 10.12: Admin Governance Dashboard & Platform Metrics

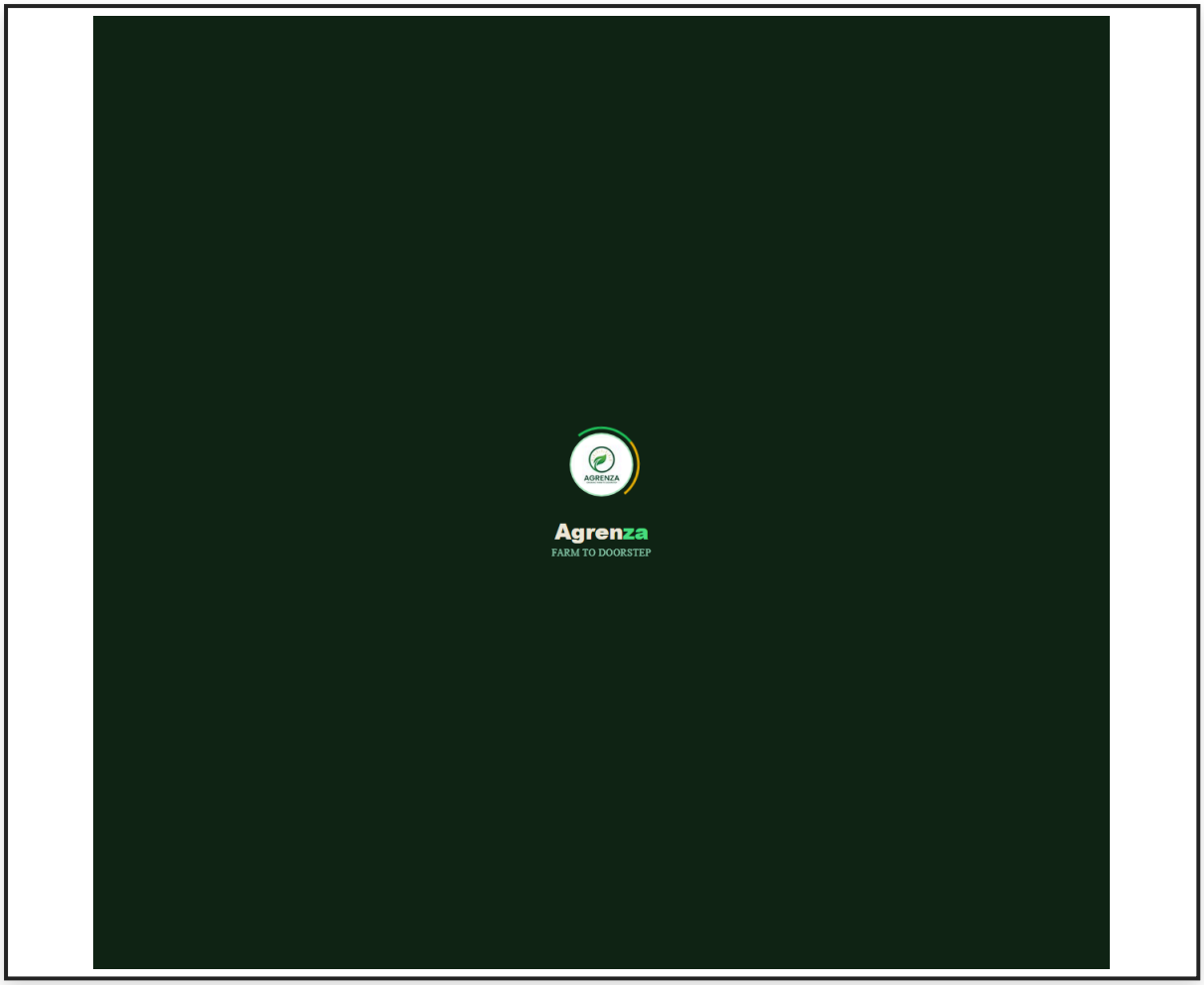


Figure 10.13: Admin Dual-Role Shared Account Auditor